A Bootstrap-Based Reliability Framework for Feature Selection in Small Medical Datasets

***Abstract*—Feature selection is difficult in small medical data sets because small sample size causes a large amount of variation in the selection of the feature subset. Even though the selection frequency is widely used for assessing the stability of feature selection, it fails to account for the uncertainty, predictive importance, and redundancy of features. This paper develops a Reliability-Based Stability Framework for selecting reliable features in small high-dimensional medical data sets. The proposed framework utilizes four complementary criteria: selection stability determined via repeated bootstrap sampling, statistical reliability of the selection frequency estimation, predictive importance, and redundancy of features. One thousand bootstrap samples are generated for determining consistency in feature selection, whereas the feature-selection strategies used are the Random Forest (RF) and Mutual Information (MI). The developed framework is validated on a differentiated thyroid cancer data set consisting of 383 patients' records and 16 clinical features, which lead to 43 potential candidate features after processing and encoding. To quantify the uncertainty in the feature-selection frequency, 95% Wilson confidence interval is calculated for each potential candidate feature, and its lower confidence bound is included in the Reliability Score. The experimental outcomes show that the proposed model offers a better-ranking of the features taking into account their reproducibility, statistical reliability, contribution to prediction, and redundancy. The features that have the highest reliability scores when RF, Response Structural Incomplete, Response Excellent, and Risk models are used include Response Structural Incomplete, Response Excellent, and Risk. On the other hand, the reliability analysis based on MI reveals that Response Structural Incomplete, T, and Response Excellent are the most reliable features.**

***Index Terms*—Feature Selection, Stability Selection, Bootstrap Resampling, Reliability Score, Uncertainty Estimation, Redundancy Analysis, Random Forest, Mutual Information, Small Medical Datasets, Differentiated Thyroid Cancer**

I. INTRODUCTION

The selection of features is important in machine learning pipelines since it helps to choose features which are informative and not redundant to decrease the complexity of the model, computation cost, and prevent the overfitting problem. This step becomes especially relevant when working with biomedical datasets where the amount of variables can be relatively high compared to the patients' number. Differentiated thyroid cancer (DTC) is the most frequently occurring thyroid cancer, and despite the good prognosis of the majority of patients following standard treatment, there is still the issue of recurrence. Thus, machine learning techniques were increasingly used for predicting the recurrence on the basis of clinical, pathological and prognostic features. Recently, one of the papers analyzed machine learning algorithms for DTC recurrence prediction based on a small-sized data set of 383 patients. SHAP-based feature importance method was used to identify the contribution of each single feature. In this research, XGBoost showed an AUC-ROC value of about 90%, whereas Response became the most important feature [1]–[5].

The other important issue with regard to feature selection is feature-selection instability. In small sample sizes, even a slight change in the training set can lead to entirely different sets of features. Several recent studies have indicated that there can be wide variations in feature-selection stability across various methods despite similar predictive ability [6]–[9]. Resampling through bootstrap offers a natural way of studying this phenomenon by repeatedly perturbing the data and estimating the frequency with which each feature is chosen [10], [11]. More recent research has looked into probabilistic methods and ensembles for gaining stability in the feature subsets selected [7], [12], [13].

But feature selection frequency does not tell the whole story. The two features having the same selection frequency could have different contribution, redundancy, and uncertainty. In medical data, correlated clinical features could result in choosing the same feature repeatedly due to its redundancy with respect to other features in the data. Thus, selecting reliable features requires taking into consideration stability, relevance, uncertainty, and redundancy [1], [8], [14].

The selection of the feature-selection algorithm also influences ranking of feature importance. Random Forest (RF) is able to take into account nonlinear dependencies and interactions and offers feature importance based on the modeling approach [15]. MI, on the other hand, estimates the dependence of features and target from a statistical standpoint independent of any specific model [16]. Because of their distinctive properties, these two measures can be considered for determining if the selected features demonstrate similar reliability regardless of selection method [13], [17], [18].

Such question becomes especially relevant in relation to the prediction of recurrence in differentiated thyroid cancer (DTC). DTC recurrence is defined by a variety of clinicopathological factors and relatively small patient sample sizes. In recent years, machine learning was used for thyroid-cancer risk stratification and recurrence prediction using feature engineering, optimization, association rule mining, and explainable machine learning [19]–[23]. Yet, the emphasis in these studies is on the prediction part rather than on feature reliability and reproducibility.

To solve this problem, this study introduces the concept of Reliability-based Stability Framework to deal with the problem of feature selection on small medical datasets. This framework

involves the assessment of the candidate features through four different components, namely: selection stability, statistical confidence, predictive contribution and redundancy. While 95% Wilson Confidence Interval is adopted to estimate the uncertainty of selection frequency, predictive contribution and redundancy offer additional support to measure the usefulness and uniqueness of the selected feature.

The framework is assessed on a DTC recurrence data set comprising 383 patients and 16 clinical variables, generating 43 candidate features through preprocessing and encoding. RF and MI techniques are independently used using 1,000 bootstrap samples to estimate feature reliability. It should be noted, therefore, that the task here is not just finding frequently selected features, but proving their statistical significance, predictivity, and non-redundancy in comparison to other features.

The main contributions of this paper include:

- Stability, confidence, predictive importance, and redundancy in one feature reliability score.
- The use of 95% Wilson confidence intervals and lower confidence bound.
- Reliability assessment using RF and MI as alternative methods for feature selection.
- Illustration of the framework on a 383 patients DTC data set using 1,000 bootstrap samples.

The rest of this paper is structured as follows. Related Work is reviewed in Section II. Section III introduces the proposed framework. The experimental setup and data set are discussed in Section IV. Section V reports the results of the study, while the discussion and limitations are presented in Section VI. Conclusion and future work are outlined in Section VII.

II. LITERATURE REVIEW

The feature selection method is extensively employed for dimension reduction, elimination of irrelevant features, and enhancing the interpretability of learning machines. Traditional feature selection techniques are commonly classified into four types: filter, wrapper, embedded, and hybrid, with the efficiency of which being largely determined by data properties, dimensionality, number of examples, and the subsequent classification problem [1], [2]. However, recent researches focus on the necessity to consider stability, redundancy, reliability, and interpretability of feature selection methods, along with predictive power [3], [4]. This issue is especially relevant for biomedical applications, where data sets often include correlated features and small number of examples.

Feature-selection stability is defined as the robustness of the chosen features in case of some changes to the data or experimental setting. Nogueira et al. [11] have created a framework for the assessment of feature-selection stability and proved that stability has to be estimated separately from the prediction. Recent research suggests that stability definition significantly affects the assessment of robustness of features in the case of data drift, missing values, and other perturbations [6]. Probability-based approaches to stability have been used for finding robust features in the context of high-dimensional gene expression data [7], whereas the latest biomarker studies combine selection frequency with the feature attribution techniques, e.g., SHAP [8], [9].

The bootstrap method offers a good way of testing selection stability since it involves sampling perturbed sets of data from the available data set. Bootstrap was first proposed by Efron [10][and has later been used in feature selection to measure the stability of the selected variables. Selection frequency is an empirical measurement and can have a lot of uncertainty especially if the size of the available samples is small. In such cases, there is need for the measurement of stability to include an estimation of uncertainty in the measurement. Nogueira et al. [11] emphasized the role of statistical inference in measuring stability

An additional promising avenue of research is ensemble and multi-selector feature selection. Using various feature-selection techniques may help avoid reliance on one algorithm and offer complementary information about feature relevance. Specifically, Xiang et al. [17] introduced a technique for improving stability based on ensemble learning, which uses both stability and prediction measures to ensure robust feature selection. Graph-based ensembles have also been suggested for biomedical data sets through modeling inter-variable dependencies and solving the problems related to high dimensionality and low sample size [18]. Recently, it has been shown using deep-learning-based feature-attribution approach that stability, predictability, and interpretability of selected features differ significantly depending on the feature selection method used [13].

Redundancy and predictive contributions are further aspects that need to be considered while assessing the reliability of features. High correlation among features might lead to multiple selections of the same features which contribute redundant information. The Redundancy Minimization and Maximum Relevance (mRMR) method, which was developed by Peng et al. [16], highlighted the significance of balancing the relevance of the features with redundancy. Redundancy continues to play an important role in recent biomedical research for the selection of the features especially when several clinical/biological variables are used to convey the same information [1], [18]. Nonetheless, many stability-based methods have only considered the aspect of consistency while neglecting others. Thyroid cancer recurrence prediction through machine learning has also gained significance. Various research works have been carried out to evaluate machine learning models for risk stratification and recurrence prediction in thyroid cancer patients [19], [20]. In one such study, Atay et al. [21] developed a hybrid approach involving association rule mining and classification for predicting recurrence of differentiated thyroid cancer. Other works have explored the use of machine learning techniques for recurrence prediction through feature engineering and development of pipelines [22], [23]; recent optimization-based methods with the use of CatBoost and SHAP have identified factors like response, risk, and N stage [24]. Another recent review has also stressed on the growing importance of machine learning in the prediction

of thyroid cancer recurrence [25]. Despite the predictive power of machine learning algorithms in various studies, their major focus has remained on modeling and feature importance rather than reproducibility and statistical significance of the selected features.

These facts are evident in the literature review; hence, there is an identified research gap for investigation. To start with, conventional feature-selection research tends to be biased towards prediction efficiency without much consideration on selection replicability [3], [4]. Secondly, stability-selection techniques tend to focus more on consistency than including predictive contributions and redundancies [6], [11], [12]. Thirdly, while ensemble learning techniques improve reliability of selectors through combination, these do not include individual uncertainties in selection probabilities [13], [17], [18]. Finally, while machine learning techniques for differentiated thyroid cancer have established important recurrence related features, the stability of such features has been largely neglected [20]–[25].

To fill the above-mentioned gaps, this research presents a Reliability-Based Stability Framework that will analyze each feature from four perspectives: selection stability, statistical confidence, predictive contribution and redundancy. As opposed to the conventional selection techniques which consider only frequency or predictive contribution, this research technique offers a multi-dimensional reliability analysis technique with repeated bootstrapping and complementary Random Forest (RF) and Mutual Information (MI) selectors.

III. PROPOSED METHODOLOGY

Overall methodology involves data preprocessing, bootstrapping, feature selection using Random Forest (RF) and Mutual Information (MI), reliability estimation, and reliable features identification. The overall procedure of the proposed framework is shown in Fig. 1. Firstly, differentiated thyroid cancer dataset is selected and then data preprocessing and feature encoding are conducted to generate the candidate feature set. Subsequently, bootstrapping with 1,000 iterations is carried out for the processed data, followed by feature selection through Random Forest (RF) and Mutual Information (MI). Next, the selected features will be evaluated based on four independent criteria including selection stability, 95% Wilson confidence, predictive contribution, and redundancy. Four criteria are further combined into one Reliability Score which is eventually applied for reliable feature identification. Through the above mentioned procedure, the proposed framework is able to evaluate the reliability of the features from multiple angles instead of simply counting the number of selections.

A. Data Preprocessing

The experiment employs a thyroid cancer recurrence dataset which comprises 383 cases and 16 clinical attributes. The target attribute is "Recurred." The categorical attributes are transformed to numerical values by means of one-hot encoding, whereas the numerical attributes are standardized. In total, after preprocessing and encoding, the initial 16 attributes generate 43 potential attributes.

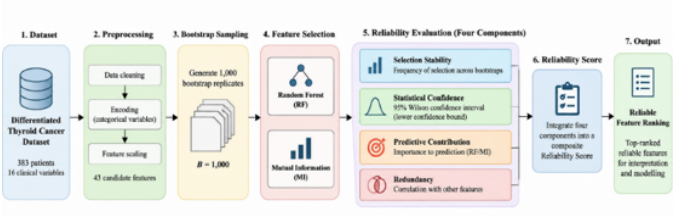


Fig. 1. Architecure of the Reliability-Based Stability Framework for feature selection in small medical datasets

In order to balance the classes, SMOTE oversampling technique is used in the training and selection phase. The preprocessing and balancing are done in a consistent way in each bootstrap round.

B. Bootstrap-Based Feature Selection

Feature selection robustness was analyzed by using 1,000 bootstrap samples, where in each bootstrap sample, a bootstrap sample is drawn and feature selection is carried out independently on the bootstrap sample.

There are two ways for feature selection:

- Random Forest (RF): This method uses the feature selection technique based on model importance and can find nonlinear relationships.
- Mutual Information (MI): This method measures the statistical relationship between the target variable and the individual features.

A feature selection matrix for each feature selection result of bootstrap replicate was created.

C. Selection Stability

Selection stability represents how consistently a feature is selected across the bootstrap samples. For feature f_j, the selection frequency is calculated as

$$S_j = \frac{1}{B}\sum_{b=1}^{B} I_j^{(b)} \tag{1}$$

where $I_j^{(B)} = 1$ if feature f_j is selected in bootstrap replicate b, and 0 otherwise. $B = 1000$ is the total number of bootstrap replicates.

A value of S_j close to 1 indicates that the feature is consistently selected, whereas a lower value indicates greater sensitivity to sampling variation.

D. Confidence and Uncertainty Estimation

Selection frequency alone does not indicate the uncertainty associated with its estimation. Therefore, a 95% Wilson confidence interval is calculated for each feature's selection frequency.

For each feature, the confidence interval is represented as

$$CI_j = [L_j, U_j] \tag{2}$$

where L_j and U_j denote the lower and upper confidence bounds, respectively. The lower confidence bound L_j is used as the conservative confidence component of the proposed Reliability Score. A higher lower bound indicates stronger statistical support for consistent feature selection.

This approach prevents a feature from being considered highly reliable solely because of a high observed selection frequency when the corresponding estimate has greater uncertainty.

E. Predictive Contribution

In addition to stability and confidence, the predictive contribution of each feature is considered. The raw contribution values obtained during the feature-selection analysis are normalized to the range [0,1]:

$$P_j = \frac{P_j^{raw} - P_{min}}{P_{max} - P_{min}} \tag{3}$$

A higher value indicates a greater relative predictive contribution of the feature.

F. Redundancy Assessment

Correlated variables can be found in medical datasets. Consequently, redundancy is part of the reliability measure. The largest absolute correlation between each variable and all other candidate variables is:

$$RD_j = max_{k \neq j} |\rho \tag{4}$$

A higher RD_j indicates greater redundancy. Consequently, the non-redundancy component is represented by $1 - RD_j$.

G. Reliability Score

The four components are combined to obtain the proposed Reliability Score:

$$R_j = \frac{S_j + L_j + P_j + (1 - RD_j)}{4} \tag{5}$$

where S_j represents selection stability, L_j represents the Wilson 95% lower confidence bound, P_j represents normalized predictive contribution, and RD_j represents feature redundancy. Equal weighting is used because the objective is to provide a balanced assessment of feature reliability without introducing additional weight parameters. A higher Reliability Score indicates that a feature is consistently selected, statistically supported, predictive, and comparatively non-redundant.

H. Reliable Feature Identification

Once the reliability score has been computed for each potential feature, the features can then be sorted on the basis of their reliability scores. Furthermore, the 95% lower confidence bound is employed to discriminate between reliably selected features and unreliable ones. For instance, a threshold reliability score based on the lower bound can be used, as $L_j \geq 0.50$, meaning that the feature has a selection probability of 50% or more.

IV. EXPERIMENTAL SETUP

A. Dataset Description

The suggested framework is tested on a recurrence of differentiated thyroid cancer dataset with 383 patient records and 16 clinical variables. Recurred is considered as a target variable, where 1 indicates that recurrence was observed within follow-up. Clinical variables cover demographic, clinical, pathological, and staging data such as age, gender, smoking, thyroid function, physical examination, adenopathy, pathology, focality, risk, T, N, M, stage, and response after treatment.

After categorical encoding and preprocessing, the initial set of 16 clinical variables produces 43 candidate features. It is necessary to provide the numeric representation of categorical clinical information as well as maintain the information needed for feature-selection analysis.

B. Experimental Configuration

For assessing the stability of feature-selection procedure, 1,000 bootstraps are produced. Random Forest (RF) and Mutual Information (MI) are independently run on each bootstrap. For each bootstrap, the selected features are recorded in a binary selection matrix.

Using the selection matrices, selection frequency, Wilson 95% confidence interval, predictive contribution, redundancy penalty, and Reliability Score are calculated. Reliability-analysis approach is independently applied to both selectors.

V. RESULTS AND DISCUSSION

A. Random Forest-Based Reliability Analysis

As shown in Fig. 2 based on the corrected B = 1000 RF model, Response_Structural_Incomplete, Response_Excellent, and Risk are among the top-ranked features having a selection frequency of 1.000 as these features have been selected in each of the 1,000 bootstrap samples. Adenopathy_No and N_N0, on the other hand, had very high selection frequencies (around 0.95), followed by N_N1b (around 0.85) and T (around 0.84). Meanwhile, Age_scaled had a comparatively moderate selection frequency around 0.72, meaning that while it had been selected in many bootstrap samples, it was less stable than the aforementioned highly ranked features. Response_Indeterminate was selected in around 56% of the bootstrapping processes, while Stage had an extremely low selection frequency of around 0.10. In short, Fig. 2 shows that through bootstrapping, one can identify many consistent features through different samples while at the same time identifying variability within less stable features. These are the selection-frequency results that would be used for further Reliability Score analysis.

B. Mutual Information-Based Reliability Analysis

From the corrected B = 1000 MI model, Fig. 3 shows the selected features based on the number of times each feature was selected during the bootstrap replicates. The Risk, Response_Excellent, Adenopathy_No, and Response_Structural_Incomplete features were selected in all

TABLE I
TOP RF FEATURES BASED ON RELIABILITY SCORE.

Rank	Feature	Stability	95% CI Lower	Predictive Contribution	Redundancy	Reliability
1	Response_Structural Incomplete	1.000	0.996	1.000	0.000	0.999
2	Response_Excellent	1.000	0.996	0.554	0.425	0.781
3	Risk	1.000	0.996	0.506	0.475	0.757
4	N_N0	0.946	0.930	0.268	0.520	0.656
5	Adenopathy_No	0.932	0.915	0.272	0.581	0.634
6	T	0.921	0.903	0.232	0.561	0.624
7	N_N1b	0.869	0.847	0.236	0.606	0.586
8	Age_scaled	0.565	0.534	0.173	0.378	0.474

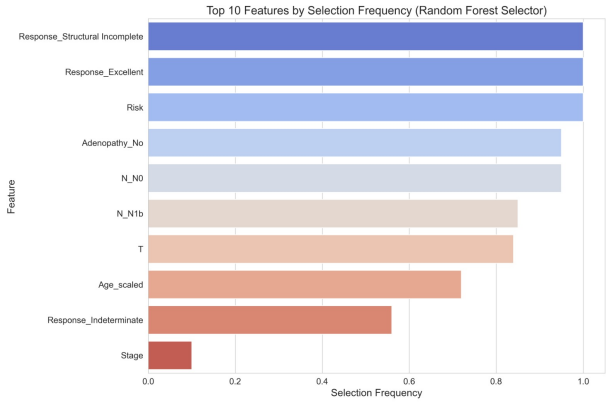


Fig. 2. Top-ranked features based on selection stability and corrected Reliability Score using Random Forest.

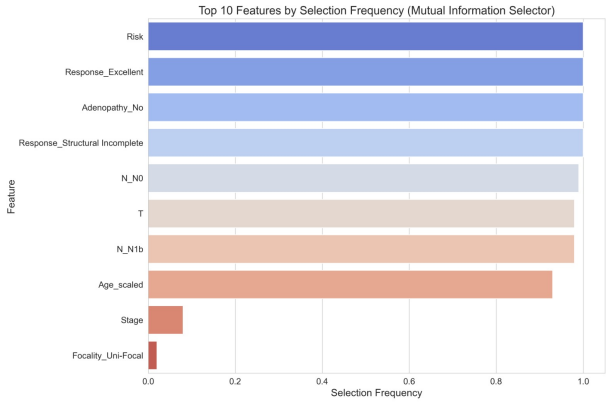


Fig. 3. Top-ranked features based on selection stability and corrected Reliability Score using Mutual Information.

1,000 bootstrap replicates, hence obtaining the selection frequency of 1.000. Other features that were found to be stable in their selection included the following features; N_N0 (0.99), T (0.98), N_N1b (0.98), and Age_scaled (0.93). On the other hand, the Stage was found to have much lower selection frequency with approximately 0.08 while Focality_Uni-Focal feature was selected in 0.02 of the bootstrap replicates. Therefore, from the above discussion, it is clear that the MI selector selects certain clinical features consistently from sample to sample such as risk, treatment response, adenopathy and staging features. However, the fact that Stage and Focality_Uni-Focal have lower selection frequencies means that the two are more sensitive to the data samples.

C. Comparison Between RF and MI

The comparison of feature Reliability Scores obtained from the two selection procedures is presented in Fig. 4. The scatter plot shows the Reliability Score produced by the RF selector on the x-axis and that produced by the MI selector on the y-axis, with the dashed diagonal representing perfect agreement. Several features are located close to the diagonal, indicating comparable reliability estimates from both selectors. In particular, Response_Structural_Incomplete, Response_Excellent, Risk, N_N0, and N_N1b show relatively consistent scores across the two methods. However, noticeable differences are observed for several features. T and Age_scaled receive considerably higher Reliability Scores from MI than

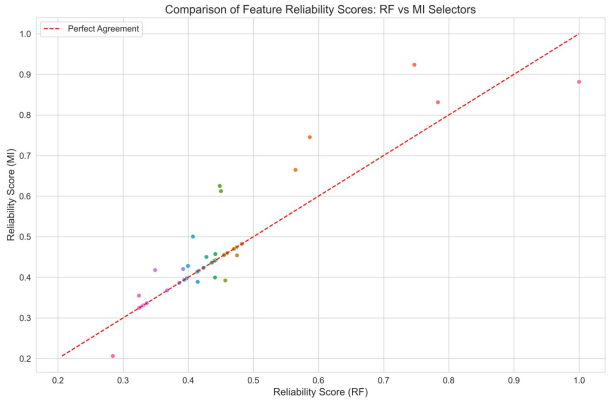


Fig. 4. Comparison of feature Reliability Scores obtained using Random Forest and Mutual Information.

from RF, while some other features also deviate from the perfect-agreement line. Conversely, the RF and MI scores for Response_Structural_Incomplete remain high across both methods, with values of approximately 0.999 and 0.896, respectively. Overall, Fig. 4 demonstrates that although the two selectors identify several features with similar reliability, their score distributions are not identical. This variation highlights the influence of the underlying selection mechanism and supports the use of RF and MI as complementary feature-selection procedures within the proposed framework.

TABLE II
TOP MI FEATURES BASED ON RELIABILITY SCORE

Rank	Feature	Stability	95% CI Lower	Predictive Contribution	Redundancy	Reliability
1	Response_Structural Incomplete	1.000	0.996	1.000	0.412	0.896
2	T	1.000	0.996	0.573	0.000	0.892
3	Response_Excellent	1.000	0.996	0.751	0.428	0.830
4	Risk	1.000	0.996	0.748	0.561	0.796
5	Adenopathy_No	1.000	0.996	0.518	0.438	0.769
6	Age_scaled	0.997	0.991	0.367	0.378	0.744
7	N_N0	1.000	0.996	0.516	0.606	0.727
8	N_N1b	1.000	0.996	0.491	0.603	0.721
9	Focality_Uni-Focal	0.890	0.869	0.283	0.371	0.668
10	Stage	0.940	0.924	0.292	0.583	0.643

D. Confidence-Based Interpretation

Another feature considered in the application of the Wilson confidence interval is the uncertainty in the value of the observed selection frequency. It can be seen from Fig. 5 that the features which were selected in all 1,000 repetitions of bootstraping have the observed selection frequency of 1.000, with the Wilson 95% lower confidence bound equal to about 0.996. This demonstrates a very high level of stability in spite of the limited number of bootstrap samples used. However, the RF feature Response_Indeterminate is chosen with the observed selection frequency of 0.510 and the lower confidence bound equal to about 0.479, while the MI feature Adenopathy_Bilateral is chosen with the observed selection frequency of 0.488 and the lower confidence bound of about 0.457. Thus, it appears that moderately stable features have higher uncertainty than the features which are selected with the same high frequency in all bootstrap samples. Fig. 5 also reveals that the features with high stability have confidence intervals located close to the observed values of selection frequency, whereas moderately stable features have wider confidence intervals.

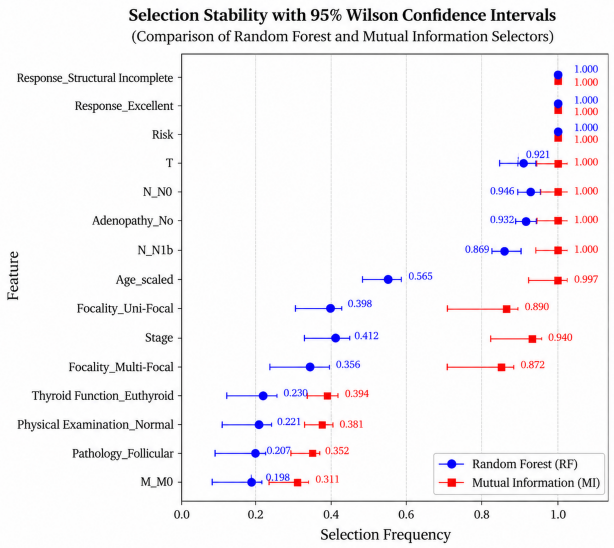


Fig. 5. Selection stability (with 95% Wilson Confidence intervals) for the top 15 features using Random Forest and Mutual Information selectors. The points indicate the observed selection frequency, and the horizontal bars represent 95% Wilson confidence intervals.

E. Reliability Score Analysis

An adaptive threshold based on the largest-gap method was considered in the original framework; however, the corrected B = 1000 experiment shows that the resulting threshold is highly dependent on the feature selector. For RF, the largest gap occurs immediately after the top-ranked feature, producing a threshold of 0.999 and retaining only one feature. In contrast, for MI, the largest gap occurs much lower in the Reliability Score distribution, resulting in a threshold of 0.136 and retaining 33 features, as summarized in Table III. The results indicate that the largest-gap approach is strongly influenced by the distribution of composite Reliability Scores. Therefore, although it can provide an adaptive threshold, it should not be considered a validated feature-selection criterion without independent predictive validation. In this study, the Reliability Score is primarily used as a feature-ranking and reliability-assessment measure. The lower Wilson confidence bound is instead considered a potential conservative criterion for identifying a reliable feature subset, which can be further validated in future predictive experiments.

TABLE III
ADAPTIVE GAP ANALYSIS

Selector	Threshold	Largest Gap	Cut Rank	Retained Features
RF	0.999	0.218	1	1
MI	0.136	0.136	33	33

VI. DISCUSSION

The experimental outcomes indicate that the proposed approach offers additional knowledge compared to classical selection frequency. RF and MI select the same informative features like response-related factors, risk, T stage, N stage, and age but provide different ranking of such features as focality and other clinical factors. The explicit computation of the Wilson confidence interval allows us to interpret selection frequency more reliably because the uncertainty in the small clinical dataset is estimated. Moreover, we see from the experimental results that stability and reliability are two different measures since high stability of features does not

mean high Reliability Scores due to predictive significance and redundancy. At the same time, the coincidence of RF and MI results proves the validity of selected features, whereas the difference proves the importance of the selection approach. In addition, the analysis with B=1000 shows the sensitivity of the largest-gap threshold to the Reliability Score distribution.

VII. LIMITATIONS

A number of limitations need to be addressed. First, the reliability study was done on one specific differentiated thyroid cancer dataset which has 383 instances; thus, the generalizability of the detected reliable features still needs to be verified. Second, the Reliability Score treats all of its four metrics equally, and different weighting schemes have not yet been investigated. Third, the current research is mainly concerned about the reliability of feature selection. The classification studies utilizing the selected feature sets have not been conducted, and, thus, no performance improvement in terms of Accuracy, F1-score, ROC-AUC, or MCC can be claimed at this point. Fourth, the adaptive largest gap threshold revealed some selector-dependency that needs to be verified prior to implementation. Lastly, although RF and MI give a different view on feature selection, more selectors such as LASSO, recursive feature elimination, or other stability selections could be included into the study.

VIII. CONCLUSION AND FUTURE WORK

This paper introduced Reliability-Based Stability Framework for assessing the reliability of features in small medical data sets. This approach is a fusion of selection stability, statistical confidence, predictive relevance, and redundancy to form a single Reliability Score. From the 1,000 bootstrap replicates performed on the differentiated thyroid cancer data set with 383 cases and 43 features, RF and MI selected several features repeatedly, among which are the response variable related features, risk, T, N, and age. Furthermore, the Wilson 95% confidence interval offers a concrete indication of the selection frequency's uncertainty, whereas the contrast between RF and MI highlights the different feature reliability patterns. Future studies will verify the selected features by nested cross-validation and classification tests, compare this approach with traditional feature selection and stability-based methods, and explore other potential improvements.

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